

National University of Computer & Emerging Sciences

MT2005 – Probability and Statistics

Homework # 05

Uniform and Exponential Distributions

1. A cloud server processes tasks at fixed 12-second intervals (e.g., at :00, :12, :24, :36, :48 seconds every minute). A user submits a task to the server at a random time uniformly distributed between :00 and :30 seconds of a minute.
 - a) What is the probability that the task is submitted in the first 10 seconds of the minute?
 - b) What is the probability that the task waits more than 8 seconds for processing?
 - c) What is the probability that the task waits less than 4 seconds for processing?
2. a) Prove that if a nonnegative continuous random variable satisfies the memoryless property,

$$P(X > s + t \mid X > s) = P(X > t), \quad \forall s, t \geq 0$$

then it must be exponentially distributed.

- b) The total number of thousands of hours that a computer server can operate before failure is modeled as an exponential random variable with a mean lifetime of 20 thousand hours. A company is considering purchasing a used server that has already been in operation for 10,000 hours. If the company buys the server, what is the probability that it will last for at least 20,000 additional hours?
Repeat the calculation under the assumption that the lifetime of the server is not exponentially distributed, but instead is uniformly distributed over the interval (0, 40) (in thousands of hours).
3. The change of variable formula for probability density functions (PDFs) is used when we want to find the distribution of a new random variable Y that is related to an existing random variable X through a transformation. If X is a continuous random variable with PDF $f(x)$, and $Y = g(X)$, where g is an invertible transformation, the PDF of Y , denoted by $h(y)$, can be found using the following formula:

$$h(y) = f(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

where $g^{-1}(y)$ is the inverse of the transformation $g(x)$.

Suppose $X \sim \text{Exponential}(\lambda)$, then the PDF of X is given by:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- a) What is the probability density function of $Y = e^{-\lambda X}$.
 - b) What kind of random variable is Y ?
4. Evidence concerning the guilt or innocence of a defendant in a criminal investigation can be summarized by the value of an exponential random variable X , whose mean μ depends on whether the defendant is guilty. If innocent, $\mu = 1$; if guilty, $\mu = 2$. The deciding judge will rule the defendant guilty if $X > c$ for some suitably chosen value of c .
 - (a) If the judge wants to be 95% certain that an innocent man will not be convicted, what should be the value of c ?
 - (b) Using the threshold found in part (b), what is the probability that a guilty defendant will be correctly convicted?